

Physics of Semiconductor Devices

ECE210: PSD



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DELHI

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(week11)



Understanding of band bending



Poisson-Boltzmann equation

Consider the n -type side of a p - n junction. We can simplify the upcoming math by assuming that:

- 1) the donor concentration is position-independent (uniform doping)
- 2) all donors are fully ionized, and
- 3) $N_D \ll N_C$ (the semiconductor is non-degenerate).

$$n = N_D = N_c e^{(E_f - E_c)/kT}$$
$$p = n_i^2 / N_D$$

Start with Poisson's equation

$$\frac{d^2 \phi(x)}{dx^2} = -\frac{\rho(x)}{\epsilon_s}$$

Recall the electrostatic potential is related to the band-edge energy by

$$q\phi(x) = -E_c(x) + \text{constant}$$

Then we can re-write Poisson's equation in terms of the conduction band energy:

$$\frac{d^2 E_c(x)}{dx^2} = \frac{q\rho(x)}{\epsilon_s}$$

Understanding of band bending

Using the stated assumptions, we can write the charge density at any x :

$$\rho(x) = q [N_D - n(x)]$$

To relate the electron concentration to the band-bending, recall that

$$n(x) = N_C \exp \left[-\frac{E_C(x) - E_F}{kT} \right]$$

Note that in the neutral region, where there is no band-bending ($x \rightarrow \infty$)

$$n(x \rightarrow \infty) = N_C \exp \left[-\frac{E_C(x \rightarrow \infty) - E_F}{kT} \right] = n_{no} = N_D$$

$$\begin{aligned} n(x) &= N_C \exp \left[-\frac{E_C(x) - E_C(x \rightarrow \infty) + E_C(x \rightarrow \infty) - E_F}{kT} \right] \\ &= N_C \exp \left[-\frac{E_C(x) - E_C(x \rightarrow \infty)}{kT} \right] \exp \left[-\frac{E_C(x \rightarrow \infty) - E_F}{kT} \right] \end{aligned}$$

$$n(x) = N_D \exp \left[-\frac{E_C(x) - E_C(x \rightarrow \infty)}{kT} \right]$$

Define a normalized “band-bending parameter”

$$\psi_n(x) = \frac{E_C(x) - E_C(x \rightarrow \infty)}{kT} \quad (\text{No units.})$$

Then

$$n(x) = N_D \exp [-\psi_n(x)]$$

and the charge density can be written

$$\rho(x) = qN_D [1 - e^{-\psi_n(x)}]$$

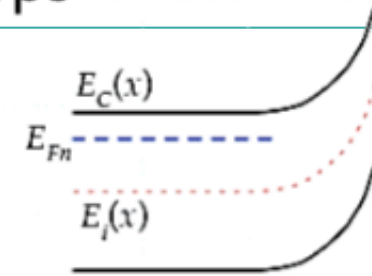
Understanding of band bending



$$n(x) = N_D \exp[-\psi_n(x)]$$

$$\rho(x) = qN_D [1 - e^{-\psi_n(x)}]$$

n-type



Upward band bending:
electrons depleted.

as the band bends up in an n-type depletion region (ψ_n is positive), the electron concentration decreases, leaving just the exposed dopant charge. However, if we have a situation where the band bends down (ψ_n is negative), then the electron concentration increases exponentially! (We call this accumulation.)

The p-type case

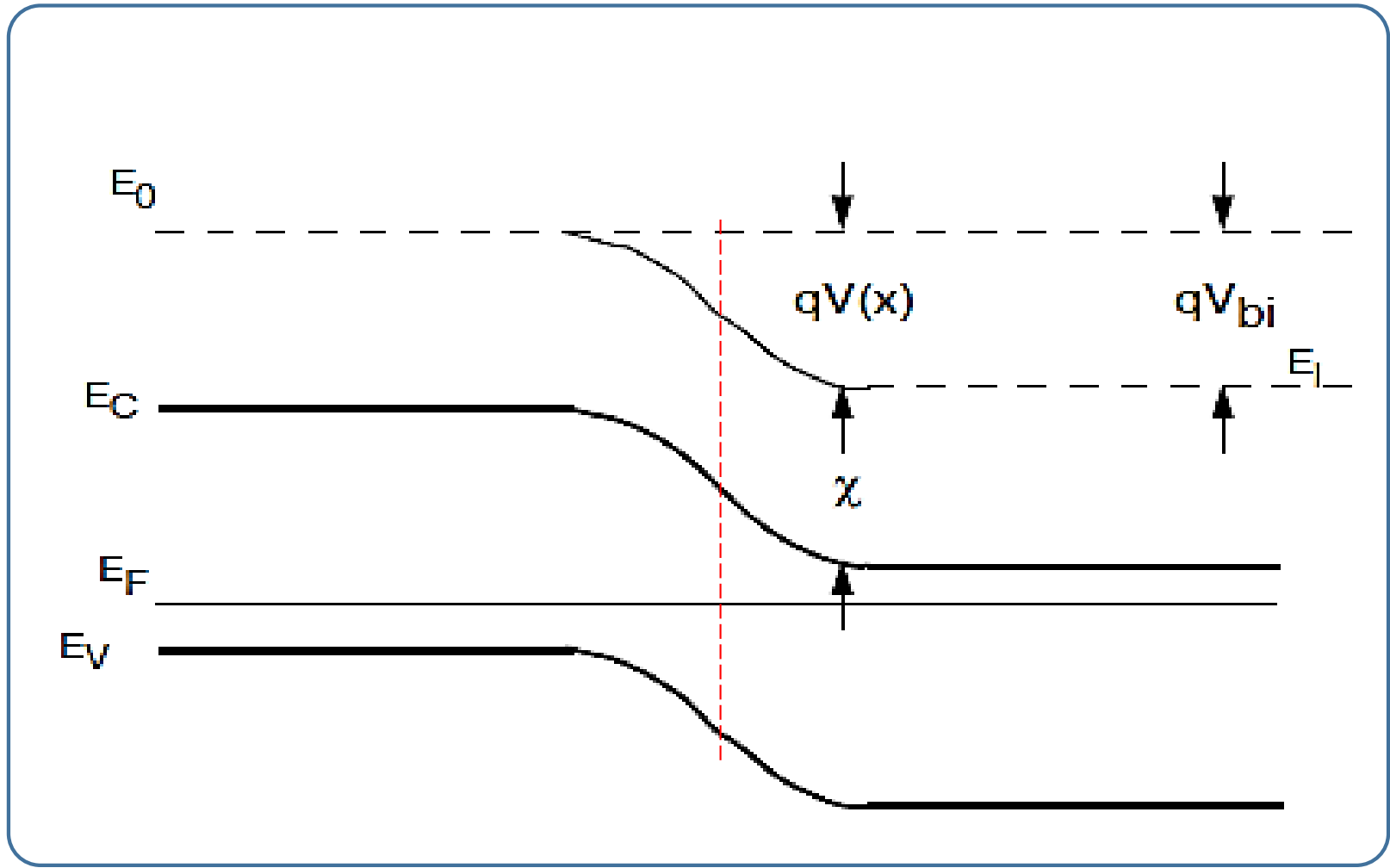
band-bending in
a p-type region.

$$\psi_p(x) = -\frac{E_V(x) - E_V(x \rightarrow -\infty)}{kT}$$

$$\rho(x) = -qN_A + p(x) = -qN_A [1 - e^{-\psi_p(x)}]$$

In the p-type case, positive ψ_p corresponds to downward band bending. This is consistent with the n-type case in that positive ψ_p represents depletion. Given that, all of the equations for the p-type case will have the same form as the n-type case.

pn diode energy band diagram



Two Governing Laws



Gauss's Law describes the relationship of charge (density) and electric field.

$$\oint_S \vec{E} \cdot d\vec{A} = \frac{1}{\epsilon} \oint_V \rho \cdot dV = \frac{Q_{encl}}{\epsilon}$$

$$\boxed{\frac{dE}{dx} = \frac{\rho}{\epsilon}}$$

$$\boxed{E(x) - E(x_0) = \frac{1}{\epsilon} \int_{x_0}^x \rho(x) dx}$$

Poisson's Equation describes the relationship between electric field distribution and electric potential

$$\boxed{\frac{d^2 \phi(x)}{dx^2} = -\frac{dE(x)}{dx} = -\frac{\rho(x)}{\epsilon}}$$

$$\boxed{\phi(x) - \phi(x_0) = \int_{x_0}^x -E(x) dx}$$

Depletion Approximation 1



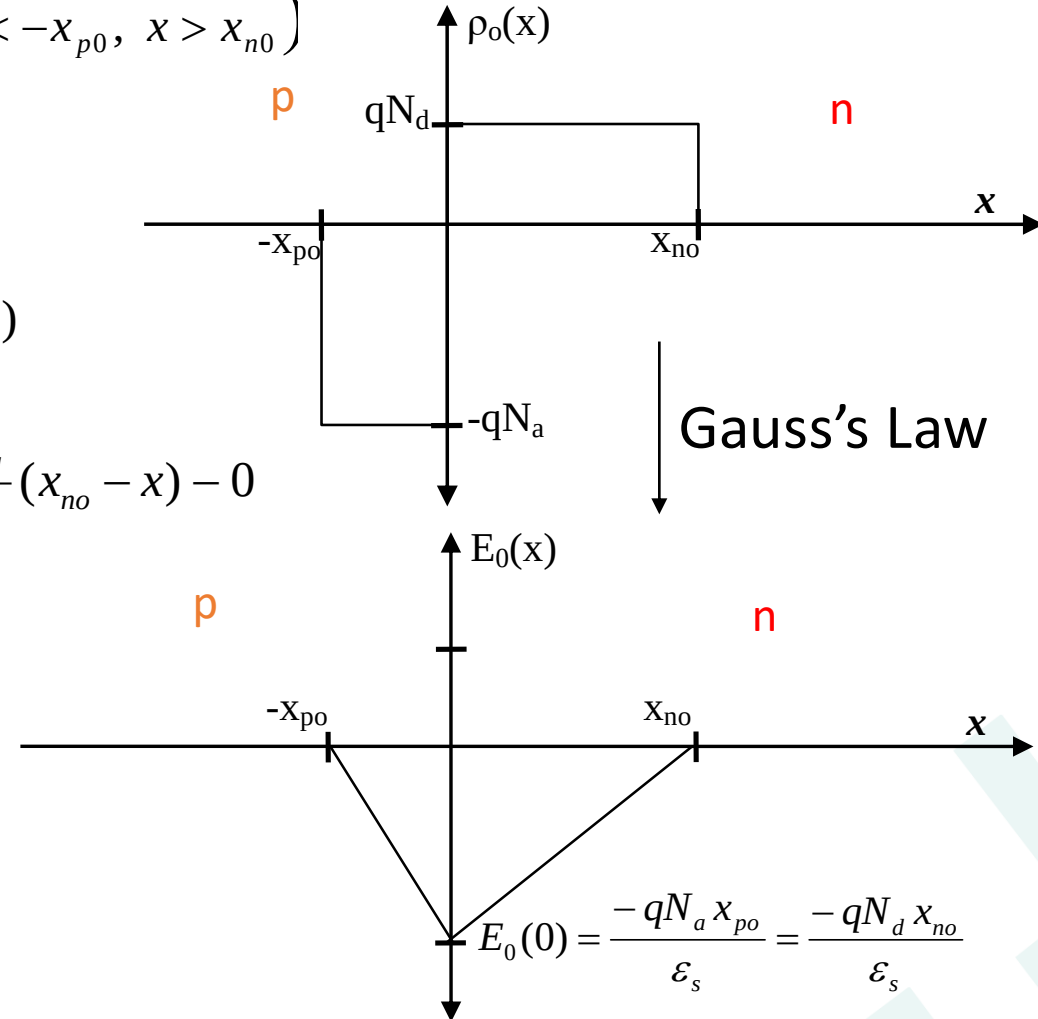
$$\rho_0(x) \approx \begin{cases} -qN_a & (-x_{p0} \leq x \leq 0) \\ qN_d & (0 \leq x \leq x_{n0}) \end{cases} \text{ and } \rho_0(x) = 0 \quad (x < -x_{p0}, x > x_{n0})$$

$$E_0(x) = \frac{-qN_a}{\epsilon_s} (x + x_{p0}) \quad (-x_{p0} < x < 0)$$

$$E_0(x) = \int_x^{x_{n0}} \frac{\rho_0(x)}{\epsilon_s} dx - E_0(x_{n0}) = \frac{-qN_d}{\epsilon_s} (x_{n0} - x) - 0$$

$$E_0(x) = \frac{qN_d}{\epsilon_s} (x - x_{n0})$$

$$(0 < x < x_{n0})$$



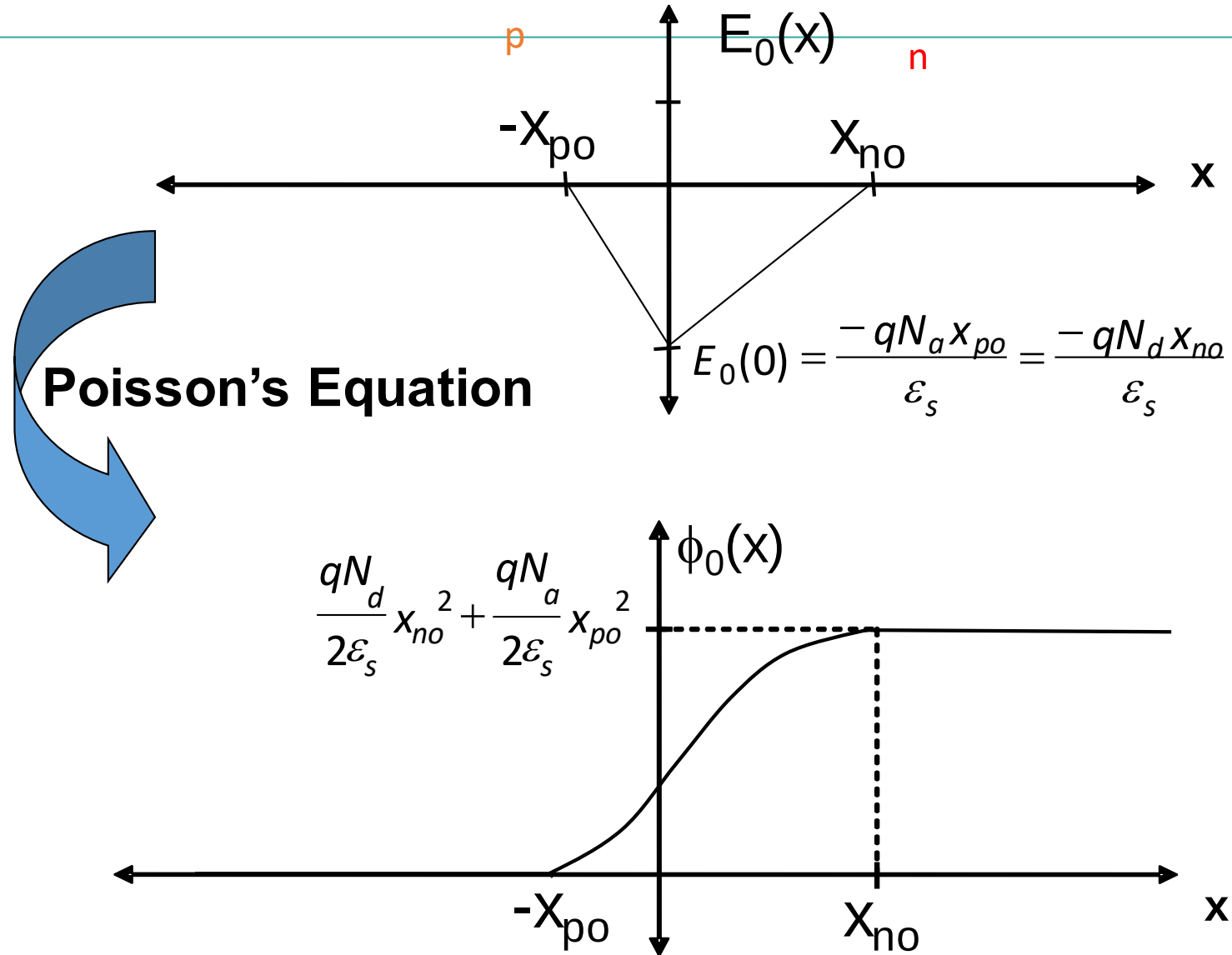
Depletion Region

$$W = \sqrt{\frac{2\epsilon_r \epsilon_o}{q} \left(\frac{N_a + N_d}{N_a N_d} \right) V_{bi}}$$

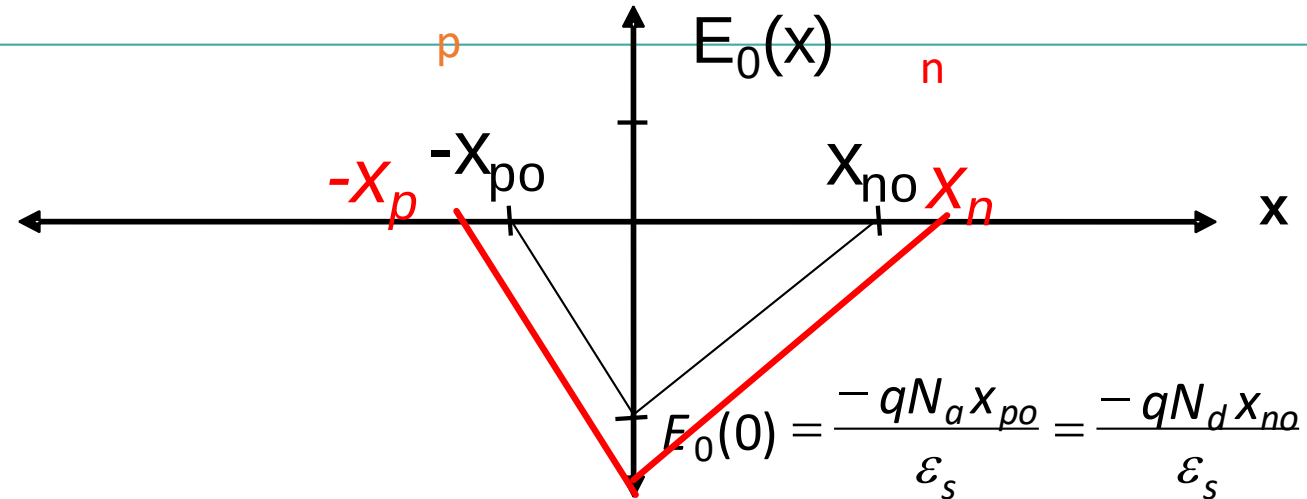
- The depletion region is inversely proportional to the doping concentration of the more lightly doped side of a p-n junction

$$W \propto \sqrt{\frac{1}{N_a}} \text{ if } N_a \ll N_d \text{ or } W \propto \sqrt{\frac{1}{N_d}} \text{ if } N_d \ll N_a$$

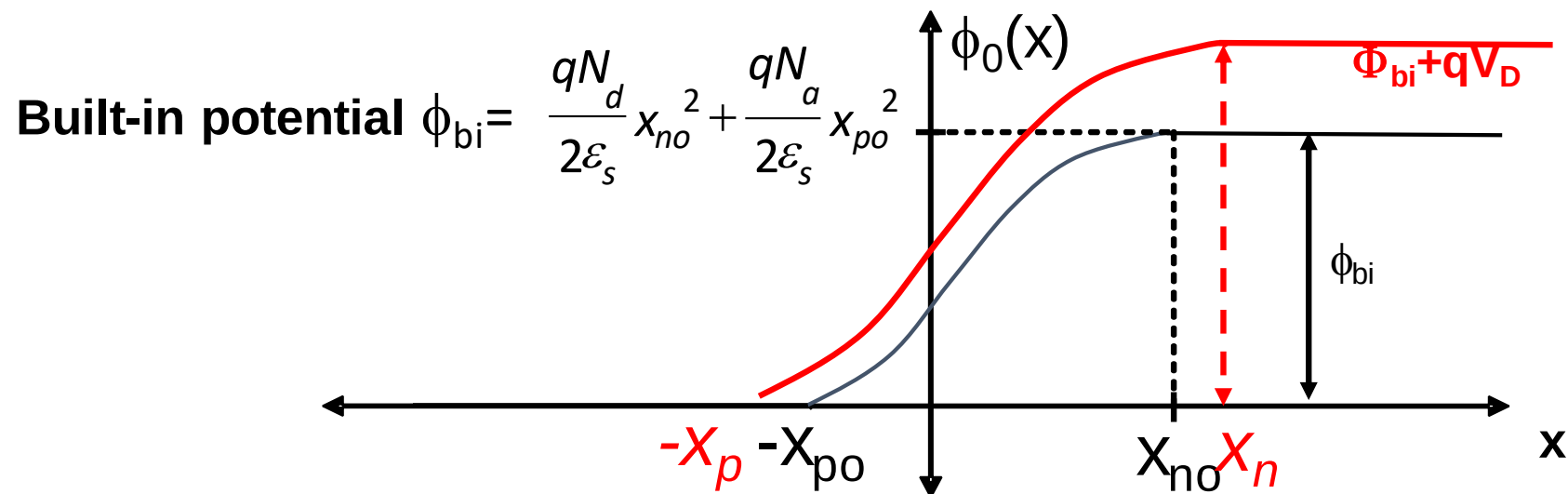
Depletion Approximation 2



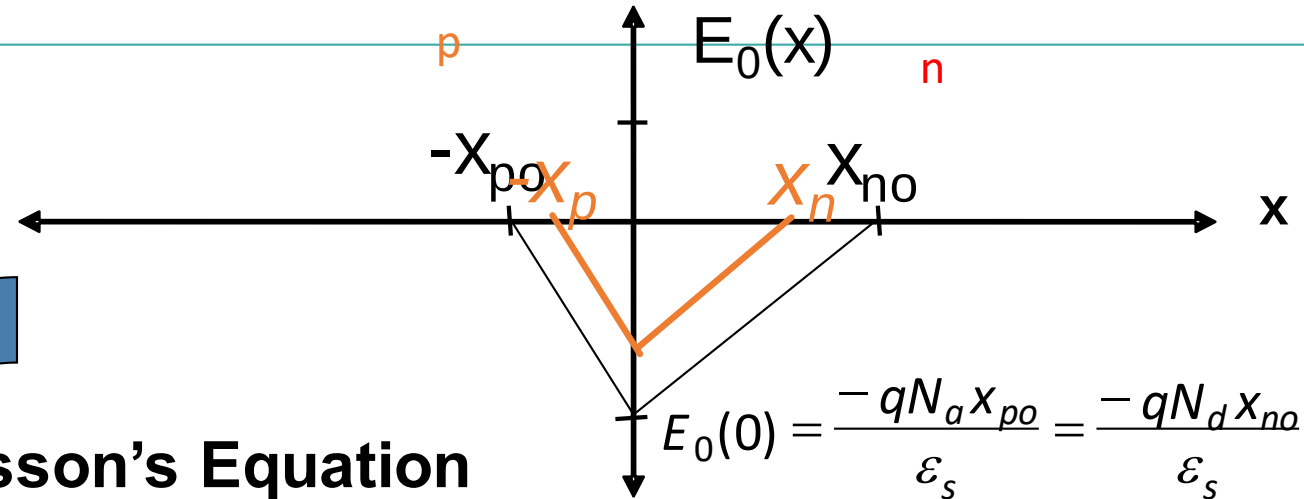
Depletion Approx. – with $V_D < 0$ reverse bias



Higher barrier and few holes in n-type lead to little current!

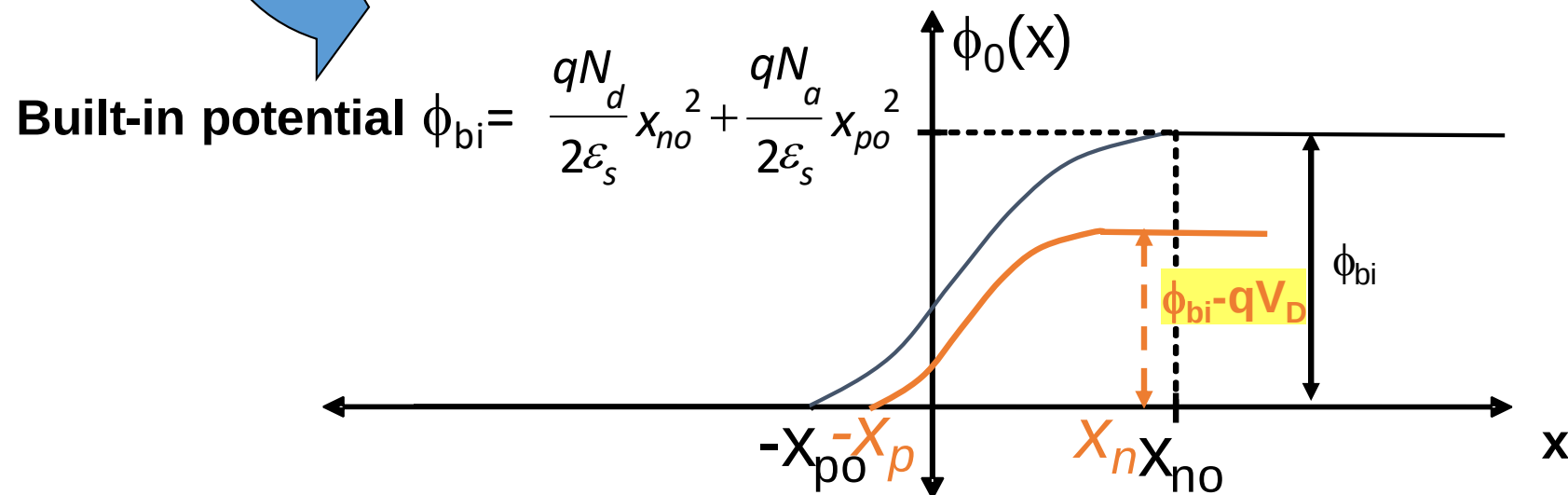


Depletion Approx. – with $V_D > 0$ forward bias



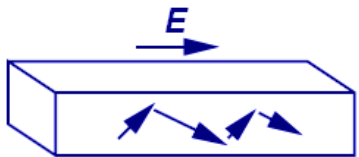
Poisson's Equation

Lower barrier and large hole (electron) density at the right places lead to large current!



pn diode energy band diagram

Drift Current



$$J_n = q \mu_n E$$

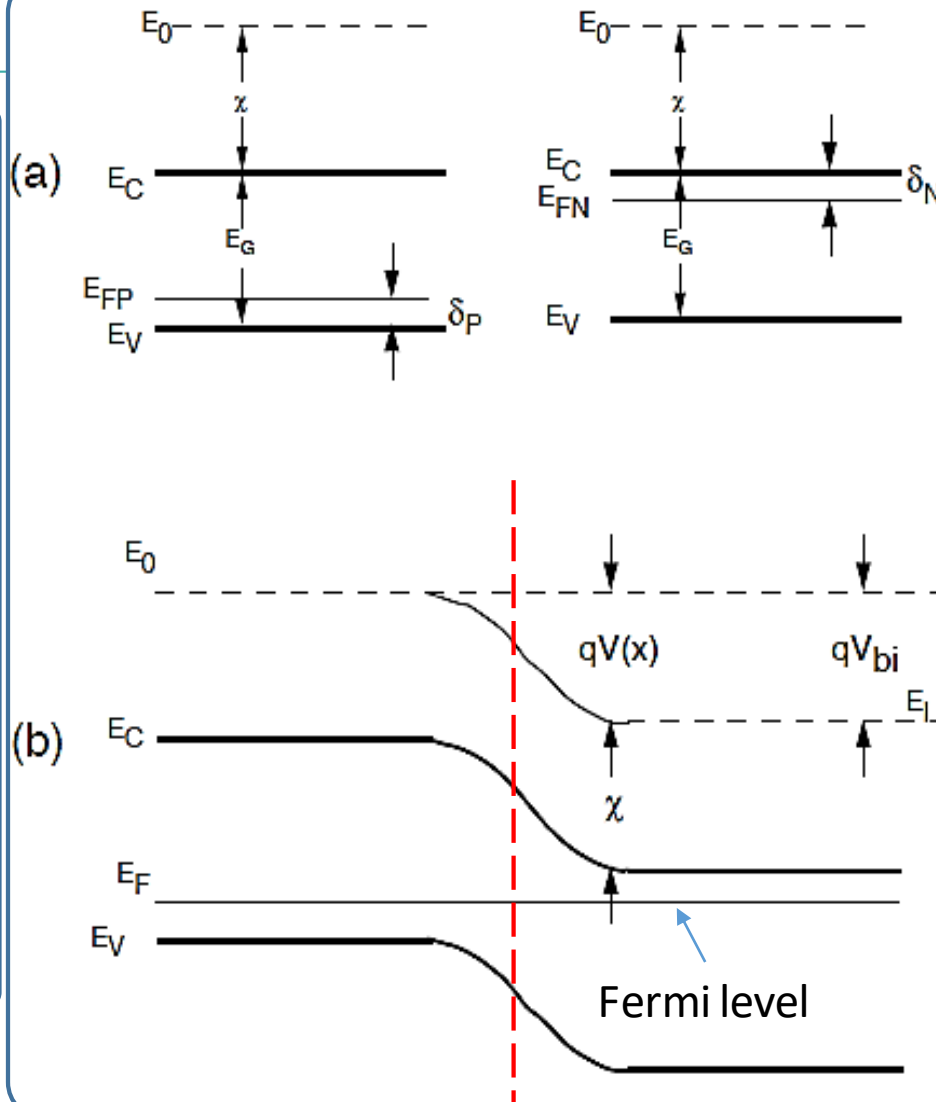
$$J_p = q \mu_p E$$

Diffusion Current

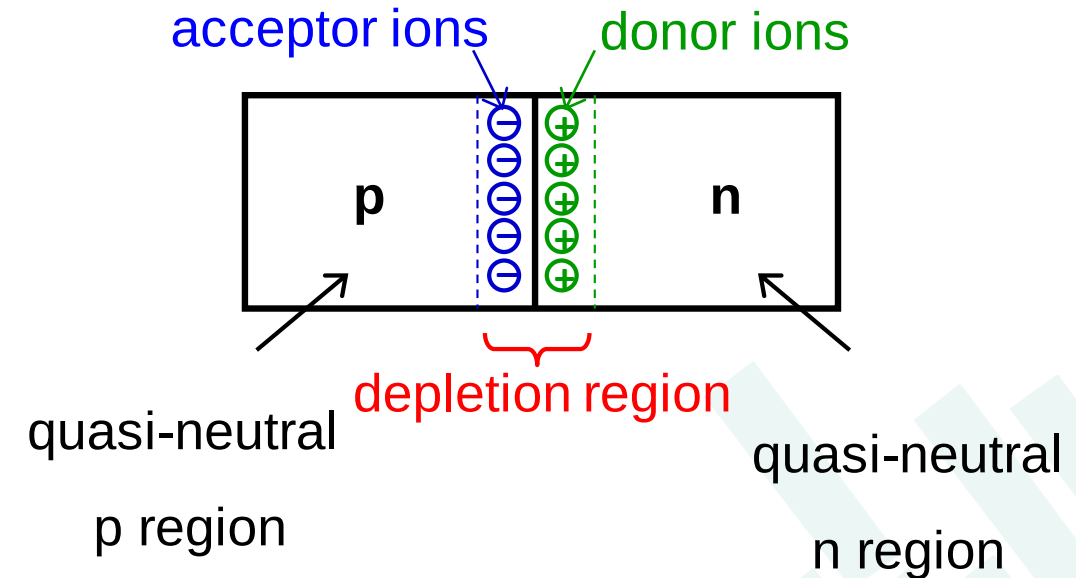


$$J_n = q D_n \frac{dn}{dx}$$

$$J_p = -q D_p \frac{dp}{dx}$$



- When the sides are “joined,” charges flow from one side to the other
- As electrons flow, they leave behind (positively charged) ionized donors
 - Donors cannot move
- Holes flow, leaving behind (negatively charged) ionized acceptors
- Separation of charges sets up electric field



In equilibrium, the drift and diffusion components of current are balanced; therefore the net current flowing across the junction is zero

Built-in Potential, V_{bi} of a pn junction



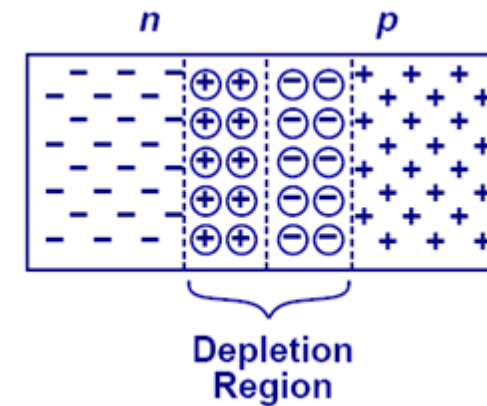
- Because of the electric field in the depletion region, there exists a potential drop across the junction:

$$qp\mu_p E = qD_p \frac{dp}{dx} \Rightarrow p\mu_p \left(-\frac{dV}{dx} \right) = D_p \frac{dp}{dx}$$

$$\Rightarrow -\mu_p \int_{x_1}^{x_2} dV = D_p \int_{p_n}^{p_p} \frac{dp}{p}$$

$$\Rightarrow V(x_1) - V(x_2) = \frac{D_p}{\mu_p} \ln \frac{p_p}{p_n} = \frac{kT}{q} \ln \left(\frac{N_A}{n_i^2 / N_D} \right)$$

$$V_{bi} = \frac{kT}{q} \ln \frac{N_A N_D}{n_i^2}$$



(Unit: Volts)